

SAM EDWARD CUTLER

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RESEARCH INTERESTS	Observational Galaxy Formation and Evolution: quenching, spatially-resolved populations, galaxy morphology, space-based data, photometric catalogs.
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EDUCATION	Ph.D. in Astronomy, University of Massachusetts Amherst (August 2025) Thesis: <i>Investigating the Origin of Low-Mass Quiescent Galaxies</i> Ph.D. Advisor: Katherine E. Whitaker M.S. in Astronomy, University of Massachusetts Amherst (September 2023) B.S. in Physics & Mathematics, University of Connecticut (May 2019) <i>Summa cum Laude</i> , Honors Scholar
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POSITIONS	Postdoctoral Scholar , Tufts University 2025 – present Graduate Research Assistant , Univ. of Massachusetts Amherst 2019 – 2025 DAWN-IRES Graduate Researcher , Cosmic Dawn Center 2022 Undergraduate Research Assistant , Univ. of Connecticut 2016 – 2019 SURF Intern , Dark Cosmology Center 2017
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ROLES	NIRCam Catalog Lead (MINERVA): <i>Created the high-quality, publicly-available NIRCam photometric catalogs for the MINERVA collaboration, leveraging over 20 photometric filters (including medium-band photometry) to achieve the best photometric redshift performance to date.</i> The Relic Co-Principal Investigator: <i>Reduced some of the deepest NIRSpec/IFU PRISM data to date, discovering young, He-enhanced globular clusters at redshift 2.5.</i> Point-spread Function Lead (UNCOVER): <i>Developed techniques to generate empirical point-spread functions and homogenization kernels for JWST NIRCam imaging, a crucial step in the photometric pipeline for UNCOVER.</i>
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SKILLS	<i>Data Handling:</i> Photometry: aperture photometry (SourceExtractor/SEP, Aperpy, Photutils), PSF matching (empirical PSF generation, Grizli, shapelets, PyPHER) Stellar Population Modeling: SED fitting (Prospector, Eazy), FSPS Morphology: profile fitting (GALFIT, statmorph, pysersic) <i>Code:</i> Python: numpy, matplotlib, scipy, astropy (expert)
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FIRST AUTHOR
PUBLICATIONS

- Cutler, Sam E., Luke Robbins, et al. (Sept. 2026). “Toward Unbreaking the Universe: MINERVA Measurements of Color Gradients in Massive Quiescent Galaxies Can Help Ease Too-early Star Formation Tensions”. In: *The Astrophysical Journal Letters* 1008.1, L11, p. L11. DOI: [10.3847/2041-8213/ae960c](https://doi.org/10.3847/2041-8213/ae960c). arXiv: [2606.02698](https://arxiv.org/abs/2606.02698) [astro-ph.GA].
- Cutler, Sam E., John R. Weaver, et al. (Nov. 2025). “The Structure and Formation Histories of Low-mass Quiescent Galaxies in the A2744 Cluster Environment”. In: *The Astrophysical Journal* 993.2, 169, p. 169. DOI: [10.3847/1538-4357/ae0629](https://doi.org/10.3847/1538-4357/ae0629). arXiv: [2504.10572](https://arxiv.org/abs/2504.10572) [astro-ph.GA].
- Cutler, Sam E., Katherine E. Whitaker, John R. Weaver, et al. (June 2024). “Two Distinct Classes of Quiescent Galaxies at Cosmic Noon Revealed by JWST PRIMER and UNCOVER”. In: *The Astrophysical Journal Letters* 967.2, L23, p. L23. DOI: [10.3847/2041-8213/ad464c](https://doi.org/10.3847/2041-8213/ad464c). arXiv: [2312.15012](https://arxiv.org/abs/2312.15012) [astro-ph.GA].
- Cutler, Sam E., Mauro Giavalisco, et al. (Mar. 2023). “The Differential Assembly History of the Centers and Outskirts of Main-sequence Galaxies at $z \sim 2.3$ ”. In: *The Astrophysical Journal* 945.2, 97, p. 97. DOI: [10.3847/1538-4357/acb5e9](https://doi.org/10.3847/1538-4357/acb5e9). arXiv: [2208.01653](https://arxiv.org/abs/2208.01653) [astro-ph.GA].
- Cutler, Sam E., Katherine E. Whitaker, Lamiya A. Mowla, et al. (Jan. 2022). “Diagnosing DASH: A Catalog of Structural Properties for the COSMOS-DASH Survey”. In: *The Astrophysical Journal* 925.1, 34, p. 34. DOI: [10.3847/1538-4357/ac341c](https://doi.org/10.3847/1538-4357/ac341c). arXiv: [2111.14848](https://arxiv.org/abs/2111.14848) [astro-ph.GA].

SELECT OTHER
PUBLICATIONS

- Whitaker, Katherine E. et al. (Apr. 2026). “Discovery of Ancient Globular Cluster Candidates in the Relic, a Quiescent Galaxy at $z = 2.5$ ”. In: *The Astrophysical Journal* 1001.1, 107, p. 107. DOI: [10.3847/1538-4357/ae5177](https://doi.org/10.3847/1538-4357/ae5177). arXiv: [2501.07627](https://arxiv.org/abs/2501.07627) [astro-ph.GA].
- Clausen, Maike et al. (Nov. 2025). “The Evolution of Half-mass Radii and Color Gradients for Young and Old Quiescent Galaxies at $0.5 < z < 3$ with JWST/PRIMER”. In: *The Astrophysical Journal* 993.1, 106, p. 106. DOI: [10.3847/1538-4357/ae03aa](https://doi.org/10.3847/1538-4357/ae03aa). arXiv: [2501.04788](https://arxiv.org/abs/2501.04788) [astro-ph.GA].
- Muzzin, Adam et al. (July 2025). “MINERVA: A NIRCам Medium Band and MIRI Imaging Survey to Unlock the Hidden Gems of the Distant Universe”. In: *arXiv e-prints*, arXiv:2507.19706, arXiv:2507.19706. DOI: [10.48550/arXiv.2507.19706](https://doi.org/10.48550/arXiv.2507.19706). arXiv: [2507.19706](https://arxiv.org/abs/2507.19706) [astro-ph.GA].
- de Graaff, Anna et al. (Feb. 2025). “Efficient formation of a massive quiescent galaxy at redshift 4.9”. In: *Nature Astronomy* 9, pp. 280–292. DOI: [10.1038/s41550-024-02424-3](https://doi.org/10.1038/s41550-024-02424-3). arXiv: [2404.05683](https://arxiv.org/abs/2404.05683) [astro-ph.GA].
- Suess, Katherine A. et al. (Nov. 2024). “Medium Bands, Mega Science: A JWST/NIRCам Medium-band Imaging Survey of A2744”. In: *The Astrophysical Journal* 976.1, 101, p. 101. DOI: [10.3847/1538-4357/ad75fe](https://doi.org/10.3847/1538-4357/ad75fe). arXiv: [2404.13132](https://arxiv.org/abs/2404.13132) [astro-ph.GA].

- Wright, Lillian et al. (Mar. 2024). “Remarkably Compact Quiescent Candidates at $3 < z < 5$ in JWST-CEERS”. In: *The Astrophysical Journal Letters* 964.1, L10, p. L10. DOI: [10.3847/2041-8213/ad2b6d](https://doi.org/10.3847/2041-8213/ad2b6d). arXiv: [2311.05394](https://arxiv.org/abs/2311.05394) [astro-ph.GA].
- Weaver, John R. et al. (Jan. 2024). “The UNCOVER Survey: A First-look HST + JWST Catalog of 60,000 Galaxies near A2744 and beyond”. In: *The Astrophysical Journal Supplement Series* 270.1, 7, p. 7. DOI: [10.3847/1538-4365/ad07e0](https://doi.org/10.3847/1538-4365/ad07e0). arXiv: [2301.02671](https://arxiv.org/abs/2301.02671) [astro-ph.GA].
- Zhang, Junkai et al. (Sept. 2023). “Dust attenuation, dust content, and geometry of star-forming galaxies”. In: *The Monthly Notices of the Royal Astronomical Society* 524.3, pp. 4128–4147. DOI: [10.1093/mnras/stad2066](https://doi.org/10.1093/mnras/stad2066). arXiv: [2307.02568](https://arxiv.org/abs/2307.02568) [astro-ph.GA].
- Park, Minjung et al. (Aug. 2023). “Rapid Quenching of Galaxies at Cosmic Noon”. In: *The Astrophysical Journal* 953.1, 119, p. 119. DOI: [10.3847/1538-4357/acd54a](https://doi.org/10.3847/1538-4357/acd54a). arXiv: [2210.03747](https://arxiv.org/abs/2210.03747) [astro-ph.GA].
- Mowla, Lamiya A. et al. (July 2022). “3D-DASH: The Widest Near-infrared Hubble Space Telescope Survey”. In: *The Astrophysical Journal* 933.2, 129, p. 129. DOI: [10.3847/1538-4357/ac71af](https://doi.org/10.3847/1538-4357/ac71af). arXiv: [2206.01156](https://arxiv.org/abs/2206.01156) [astro-ph.GA].

Not listed: Twenty publications in *The Astrophysical Journal*, seven publications in *The Astrophysical Journal Letters*, two publications in *The Astrophysical Journal Supplement Series*, two publications in *The Monthly Notices of the Royal Astronomical Society*, two publications in *Nature*, one publication in *Astronomy and Astrophysics*, and eight submitted publications.

In total: Five first author publications, 57 total publications.

SUCCESSFUL
GRANTS &
PROPOSALS

- Hubble Space Telescope Cycle 32 Medium Program GO-17730 “Fulfilling the UV Legacy of the Hubble and Webb Deep Public Frontier Field” (72 orbits; Co-I; July 2024 – present)
- James Webb Space Telescope Cycle 3 Small Program GO-6405 “Clumpy Relics: The First Spectroscopic Confirmation of Globular Clusters at $z \sim 3$ ” (20.3 hours; PI; November 2024 – present)
- Hubble Space Telescope Cycle 28 Large Treasury Program GO-16259 “3D-DASH: A Wide Field WFC3/IR Survey of COSMOS” (259 orbits; Co-I; May 2021 – April 2024)

SELECT
PRESENTATIONS
AND TALKS

MIT/Kavli Institute Monday Afternoon Talk, Cambridge, MA, USA
Invited talk — March, 2026
“*The Structure and Formation Histories of Low-Mass Quiescent Galaxies in Overdense Environments*”

Santa Cruz Galaxy Workshop, Santa Cruz, CA, USA
Poster presentation — August, 2025

“The Structure and Formation Histories of Low-Mass Quiescent Galaxies in Abell 2744”

University of California Riverside Aurora Webinar, Virtual

Invited talk — March, 2025

“The Structure and Formation Histories of Low-Mass Quiescent Galaxies in Abell 2744”

Tufts Astronomy Seminar, Medford, MA, USA

Invited talk — February, 2025

“The Structure and Formation Histories of Low-Mass Quiescent Galaxies in Abell 2744”

American Astronomical Society 245th Meeting, Nat’l Harbor, MD, USA

Contributed talk — January, 2025

“The Structure and Formation Histories of Low-Mass Quiescent Galaxies in Abell 2744”

Tinsley Workshop, New Haven, CT, USA

Contributed talk — October, 2024

“The JWST Perspective of Low-Mass Quenching”

Extreme Galaxies Conference, Reykjavik, IS

Poster presentation — June, 2024

“Two Distinct Classes of Quiescent Galaxies Revealed by Sizes and Morphologies at Cosmic Noon in JWST PRIMER and UNCOVER”

American Astronomical Society 243rd Meeting, New Orleans, LA, USA

Contributed talk — January, 2024

“Low-Mass Quiescent Galaxy Sizes in the JWST PRIMER and UNCOVER Treasury Programs”

JWST First Light Conference, Cambridge, MA, USA

Poster presentation — June, 2023

“Low Masses, Small Sizes, More Excitement: Preliminary Low-Mass Quiescent Galaxy Sizes from JWST PRIMER and UNCOVER”

American Astronomical Society 241st Meeting, Seattle, WA, USA

Contributed talk — January, 2023

“Measuring Star-Formation Histories at $z \sim 2$ with a Semi-Resolved Approach”

COSMOS Collaboration Meeting, Paris, FR

Contributed talk — July, 2022

“Galaxy Structural Properties and Star-Formation Histories with 3D-DASH”

MEDIA & PRESS RELEASES • News Article, **The Remarkable Hubble Space Telescope: 35 Years Observing the Unknown in Space** — May, 2025

- *News Article*, **Back to the Big Bang: UMass astronomers help to bring to life Webb telescope's latest images of outer space** — February, 2023
- *News Article*, **How UMass Astronomers Helped the Hubble Space Telescope Take the Widest Photo of the Universe Ever** — June, 2022

TEACHING	<i>Lecturer, UMass Amherst</i>	
	UMass Precollege Summer Astronomy Course	2020–2024
	Summer course for high school students with lab and lecture components	
	<i>Graduate Researcher, Cosmic Dawn Center</i>	
	DAWN-IRES Career Skills Seminars	Summer 2022
	Undergraduate seminars	
	<i>Teaching Assistant, UMass Amherst</i>	
	Writing About Astronomy	Spring 2021
	Undergraduate course for astronomy majors	
	The Solar System	Fall 2020 – Spring 2021
	Undergraduate course with lab component for non-majors	
	<i>Student Supervision</i>	
	Graduate: <i>Tufts</i> — Luke Robbins, Veronica Pratt	
	Undergraduate: <i>UMass Amherst</i> — Leonardo Drake, Lilian Wright, Ayesha Abdullah; <i>Cosmic Dawn Center</i> — Allan Vanzandt, Eric Rumsfeld, Hanga Andras-Letanovszky, Rebeca Reyes Carrión, Lauren Elicker	
	High School: Avery Minter	
OUTREACH	Astronomy on Tap – Western Mass: <i>Co-led the creation of the Western Massachusetts chapter of Astronomy on Tap, including finding a venue, organizing shows, managing the group's social media, hosting, and presenting at shows.</i>	
	SPARK Camp: <i>Held stargazing sessions and promoted STEM careers for SPARK, a youth camp for girls interested in STEM majors, in Summer 2018 and 2019.</i>	
HONORS & AWARDS	First MINERVA Paper, MINERVA Collaboration	2026
	MA Space Grant Grad. Research Fellowship, NASA/MASGC	2021
	Best Undergraduate Poster, Univ. of Connecticut	2018
	Babbidge Scholar, Univ. of Connecticut	2016, 2018
	CT Space Grant Undergrad. Research Fellowship, NASA/CTSGC	2018
	Michael Cantara Undergrad. Research Award, Univ. of Connecticut	2017
	New England Scholar, Univ. of Connecticut	2017